

Theme 3 Definitions

Saturday, 08 June 2024 02:15

3.7 Hyperbolic Functions

Definition of the Hyperbolic Functions

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

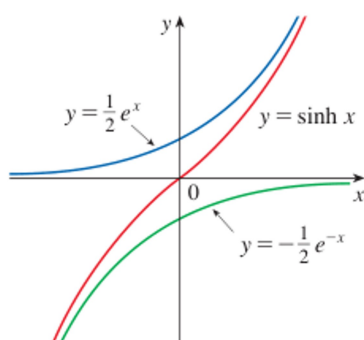


FIGURE 1

$$y = \sinh x = \frac{1}{2}e^x - \frac{1}{2}e^{-x}$$

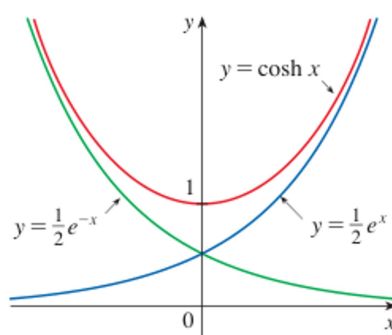


FIGURE 2

$$y = \cosh x = \frac{1}{2}e^x + \frac{1}{2}e^{-x}$$

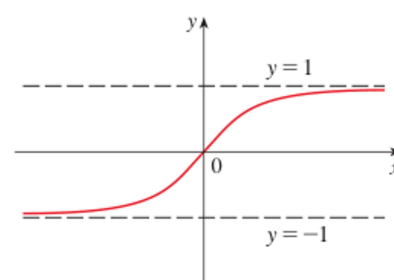


FIGURE 3

$$y = \tanh x$$

Hyperbolic Identities

$$\sinh(-x) = -\sinh x$$

$$\cosh(-x) = \cosh x$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$$